

```
! pip install tf-keras
```

```
Requirement already satisfied: tf-keras in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: tensorflow<2.21,>=2.20 in /usr/local/lib/python3.12/dist-packages (from tf-keras) (2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
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Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras) (26.0)
Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras) (75.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras) (1.16.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
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Requirement already satisfied: tensorboard~=2.20.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
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Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
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Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf-keras)
```

```
from sentence_transformers import SentenceTransformer
from sentence_transformers.util import paraphrase_mining
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
modules.json: 100% 349/349 [00:00<00:00, 5.64kB/s]
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 3.11kB/s]
README.md: 100% 10.5k/10.5k [00:00<00:00, 599kB/s]
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 2.76kB/s]
config.json: 100% 612/612 [00:00<00:00, 24.6kB/s]
model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 855MB/s]
Loading weights: 100% 103/103 [00:00<00:00, 2753.12it/s]
tokenizer_config.json: 100% 350/350 [00:00<00:00, 30.8kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 6.83MB/s]
tokenizer.json: 100% 466k/466k [00:00<00:00, 20.8MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 9.50kB/s]
config.json: 100% 190/190 [00:00<00:00, 16.7kB/s]
```

```
import sentence_transformers.util as util
s1 = "The cat is on the mat"
s2 = "A cat is sitting on a mat"
emb1=model.encode(s1)
emb2=model.encode(s2)
```

```
sim_score=util.cos_sim(emb1,emb2)
print(sim_score)
```

```
tensor([[0.9068]])
```

```
sentences = [
    "The cat sits outside",
    "A man is playing guitar",
    "I love pasta",
    "The new movie is awesome",
    "The cat plays in the garden",
    "A woman watches TV",
    "The new movie is so great",
    "Do you like pizza?",
    "She enjoys reading books",
    "He is walking his dog",
    "The weather is nice today",
    "Let's go to the beach",
    "I need a cup of coffee",
    "They are planning a trip",
    "She cooks delicious meals",
    "I am learning to play the piano",
    "He works at a tech company",
    "This book is very interesting",
    "The sun is shining brightly",
    "It's raining outside",
    "The children are playing soccer",
    "He is fixing the car",
    "She writes beautiful poetry",
    "The flowers are blooming",
    "I enjoy hiking in the mountains",
    "He is swimming in the pool",
    "We are watching a comedy show",
    "The dog is barking loudly",
    "She is painting a landscape",
    "He likes to play video games",
    "I am studying for my exams",
    "They are building a new house",
    "The coffee shop is busy",
    "He listens to classical music",
    "She is designing a website",
    "I bought a new laptop",
    "They are baking cookies",
    "The bird sings in the morning",
    "He loves going to concerts",
    "She practices yoga every day",
    "The phone is ringing",
    "I need to finish this project",
    "The restaurant has great food",
    "He collects rare coins",
    "She enjoys watching documentaries",
    "They are exploring the city",
    "The laptop is charging",
    "He is training for a marathon",
    "I meditate every morning",
    "The bakery sells fresh bread",
    "She is learning French"
]
```

```
sentences[4]
```

```
'The cat plays in the garden'
```

```
paraphrases = util.paraphrase_mining(model, sentences)
paraphrases
```

```
[0.025358058512210846, 10, 44],
[0.025195976719260216, 18, 49],
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[0.024476010352373123, 19, 36],
[0.023935552686452866, 40, 44],
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[0.022573497146368027, 24, 36],
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[0.01556527428328991, 14, 47],
[0.015424627810716629, 32, 39],
[0.01530206948518753, 8, 13],
...]
```

```
for item in paraphrases[:20]:
    score, i, j = item
    print("{} \t\t\t\t {} \t\t\t\t Score: {:.4f}".format(sentences[i], sentences[j], score))
```

The new movie is awesome	The new movie is so great	Score: 0.8939
The cat sits outside	The cat plays in the garden	Score: 0.6788
She enjoys reading books	She enjoys watching documentaries	Score: 0.6314
They are planning a trip	They are exploring the city	Score: 0.5844
He listens to classical music	He loves going to concerts	Score: 0.5794
I need a cup of coffee	The coffee shop is busy	Score: 0.5284
He likes to play video games	He loves going to concerts	Score: 0.5189
I bought a new laptop	The laptop is charging	Score: 0.5188
I love pasta	Do you like pizza?	Score: 0.5096
She enjoys reading books	She is learning French	Score: 0.5038
She cooks delicious meals	The restaurant has great food	Score: 0.5014
He likes to play video games	He listens to classical music	Score: 0.4832
I am learning to play the piano	I am studying for my exams	Score: 0.4814
She writes beautiful poetry	She is painting a landscape	Score: 0.4789
The weather is nice today	The sun is shining brightly	Score: 0.4751
A woman watches TV	She enjoys watching documentaries	Score: 0.4729
She cooks delicious meals	She writes beautiful poetry	Score: 0.4727
She is painting a landscape	She is designing a website	Score: 0.4725
They are baking cookies	The bakery sells fresh bread	Score: 0.4682
She enjoys reading books	She writes beautiful poetry	Score: 0.4526

```
model = SentenceTransformer("distiluse-base-multilingual-cased-v1")
```

```
modules.json: 100% 341/341 [00:00<00:00, 10.4kB/s]
config_sentence_transformers.json: 100% 122/122 [00:00<00:00, 2.54kB/s]
README.md: 100% 2.24k/2.24k [00:00<00:00, 148kB/s]
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 2.71kB/s]
config.json: 100% 556/556 [00:00<00:00, 41.3kB/s]
model.safetensors: 100% 539M/539M [00:10<00:00, 53.7MB/s]
Loading weights: 100% 100/100 [00:00<00:00, 2823.06it/s]
tokenizer_config.json: 100% 452/452 [00:00<00:00, 35.3kB/s]
vocab.txt: 100% 996k/996k [00:00<00:00, 15.6MB/s]
tokenizer.json: 100% 1.96M/1.96M [00:00<00:00, 37.2MB/s]
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config.json: 100% 190/190 [00:00<00:00, 16.7kB/s]
config.json: 100% 114/114 [00:00<00:00, 11.0kB/s]
2_Dense/model.safetensors: 100% 1.58M/1.58M [00:00<00:00, 3.61MB/s]
```

```
corpus1 = ["The stock market saw a big drop",
           "I want to eat paneer steam momo",
           "I like to eat white sauce pasta",
           "today, I ate panca"]
corpus_emb = model.encode(corpus1)
```

```
cosine_similarity = util.pytorch_cos_sim(corpus_emb, corpus_emb)
avg_sim = cosine_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()
outlier_index
print(f"outlier sentence: {corpus1[outlier_index]} (AVG similarity: {avg_sim[outlier_index]:.4f})")
```

```
outlier sentence: today, I ate pancake (AVG similarity: 0.3447)
```

```
corpus2= ["The cat chased a mouse across the yard.",
           "The kitten slept peacefully on the sofa.",
           "The stray cat climbed a tall tree.",
           "The airplane landed safely at the airport."]
corpus_emb = model.encode(corpus2)
```

```
cosine_similarity = util.pytorch_cos_sim(corpus_emb, corpus_emb)
avg_sim = cosine_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()
outlier_index
print(f"outlier sentence: {corpus2[outlier_index]} (AVG similarity: {avg_sim[outlier_index]:.4f})")
```

```
outlier sentence: The airplane landed safely at the airport. (AVG similarity: 0.3447)
```

```
corpus3=["I drank a glass of orange juice.",
         "She peeled a ripe banana.",
         "They bought fresh apples from the market.",
         "The teacher explained the math lesson."]
corpus_emb = model.encode(corpus3)
```

```
cosine_similarity = util.pytorch_cos_sim(corpus_emb, corpus_emb)
avg_sim = cosine_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()
outlier_index
print(f"outlier sentence: {corpus3[outlier_index]} (AVG similarity: {avg_sim[outlier_index]:.4f})")
```

```
outlier sentence: The teacher explained the math lesson. (AVG similarity: 0.3255)
```

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